

Customer Retention & Churn Analysis

Telco Customer Churn Dashboard & Insights Report

Data Science & Analytics Internship — Task 2 Submission

Dataset: Telco Customer Churn Dataset (Kaggle) | Tool: Microsoft Power BI

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Executive Summary

This report analyzes customer churn behavior for a telecommunications company using a dataset of 7,043 customers. The overall churn rate stands at 26.54%, meaning more than 1 in 4 customers leaves the service — a rate that, if unchecked, represents a significant and sustained revenue leak. Retention rate is 73.46%, meaning the majority of customers do stay, but the segments most likely to leave are clearly identifiable and largely preventable.

Four factors consistently predict which customers are most likely to churn: their contract type, how long they have been a customer, which internet service they use, and how they pay. Customers on month-to-month contracts, using fiber optic internet, paying via electronic check, and in their first 12 months of tenure churn at dramatically higher rates than any other segment. These four variables alone define the highest-risk customer profile in this dataset, and they form the basis of every recommendation in this report.

About This Analysis

Objective: to analyze customer and subscription data to understand why customers leave, which segments are most at risk, how long customers typically stay active, and what actions the business can take to improve retention.

- Dataset: Telco Customer Churn Dataset (Kaggle) — 7,043 customer records with 21 fields including demographics, services subscribed, contract type, payment method, monthly charges, total charges, tenure (months active), and churn status (Yes/No).
- Tool used: Microsoft Power BI Desktop — Power Query for data cleaning, DAX measures for KPI calculations, and an interactive retention dashboard with slicers.
- Data cleaning performed: fixed TotalCharges column (stored as text due to blank values for zero-tenure customers — replaced nulls with 0 and converted to Decimal Number); converted SeniorCitizen from 0/1 to Yes/No text; removed duplicate customerIDs; removed blank rows; renamed query to TelcoChurn for clean DAX referencing.
- Calculated columns added: ChurnFlag (1/0 numeric version of Churn for DAX math), SignupDate (back-calculated from tenure using EDATE), CohortMonth (MMM-YYYY label from SignupDate), TenureGroup (0–12 / 13–24 / 25–48 / 49+ months).

- DAX measures created: Total Customers, Churned Customers, Churn Rate %, Retention Rate %, Avg Tenure Months, Avg Monthly Charges.

Data Loading & Preparation

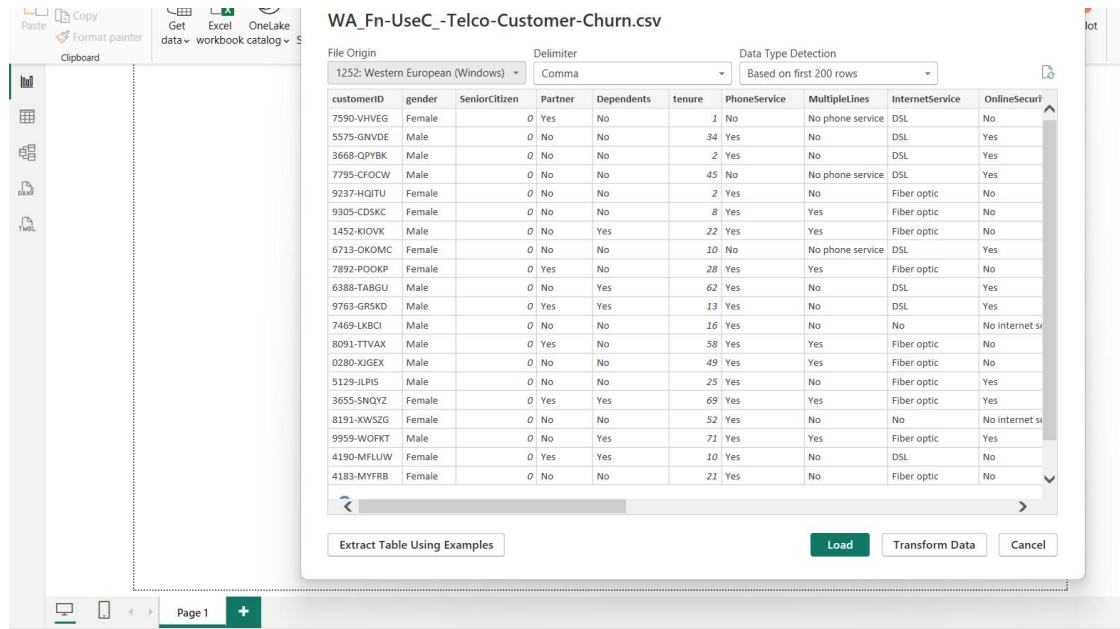


Figure 1. Data preview in Power BI before loading — 21 columns, comma-delimited, correct headers confirmed.

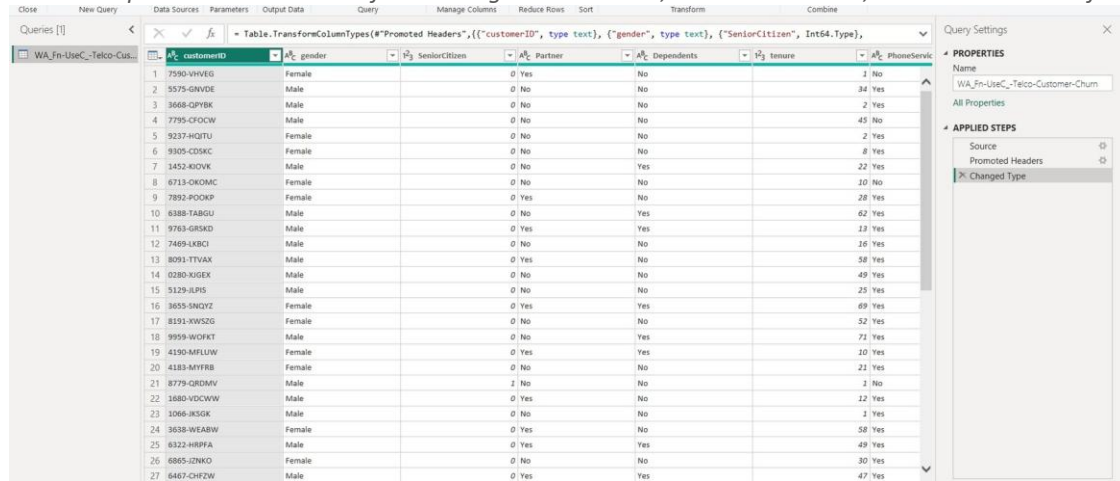


Figure 2. Power Query Editor showing the raw data and Applied Steps pipeline during cleaning.

Figure 3. Data view showing all 4 calculated columns added (ChurnFlag, SignupDate, CohortMonth, TenureGroup) alongside original data.

The screenshot shows a data table with columns: gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, and TechSupport. The table contains 20 rows of data. The calculated columns are Churn Rate, Churned Customers, Retention Rate, Total Customers, etc.

gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport
Female	Yes	Yes	No	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	No	Yes	No	7	Yes	No	Fiber optic	No	No	No	No
Female	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	Yes	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	Yes	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
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Female	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	No	No	No	7	Yes	No	Fiber optic	No	No	No	No
Male	Yes	No	No	7	Yes	No	Fiber optic	No	No	No	No
Female	Yes	Yes	Yes	7	Yes	No	Fiber optic	No	No	No	No
Male	No	No	No	7	Yes	No	Fiber optic	No	No	No	No

Figure 4. All 6 DAX measures created and visible in the Fields panel (Churn Rate, Churned Customers, Retention Rate, Total Customers, etc.).

Key Performance Indicators

TOTAL CUSTOMERS 7,043	CHURNED CUSTOMERS 1,869	CHURN RATE 26.54%	RETENTION RATE 73.46%
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The average customer tenure is 32.4 months, and the average monthly charge is \$64.76. These baseline figures set the context: the business retains most customers, but the 26.54% churn

rate means roughly 1,869 customers have already left, and each one represents lost recurring monthly revenue that is expensive to replace through new acquisition.

Dashboard Overview

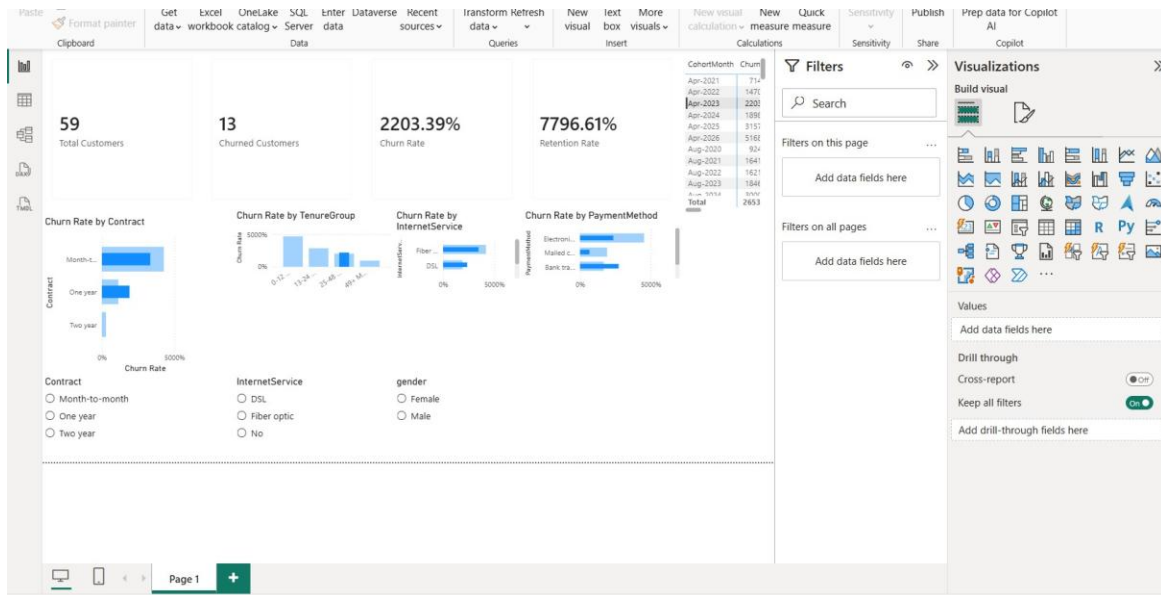


Figure 5. Full Power BI dashboard — KPI cards, churn by contract/tenure/internet service/payment method, cohort table, and interactive slicers.

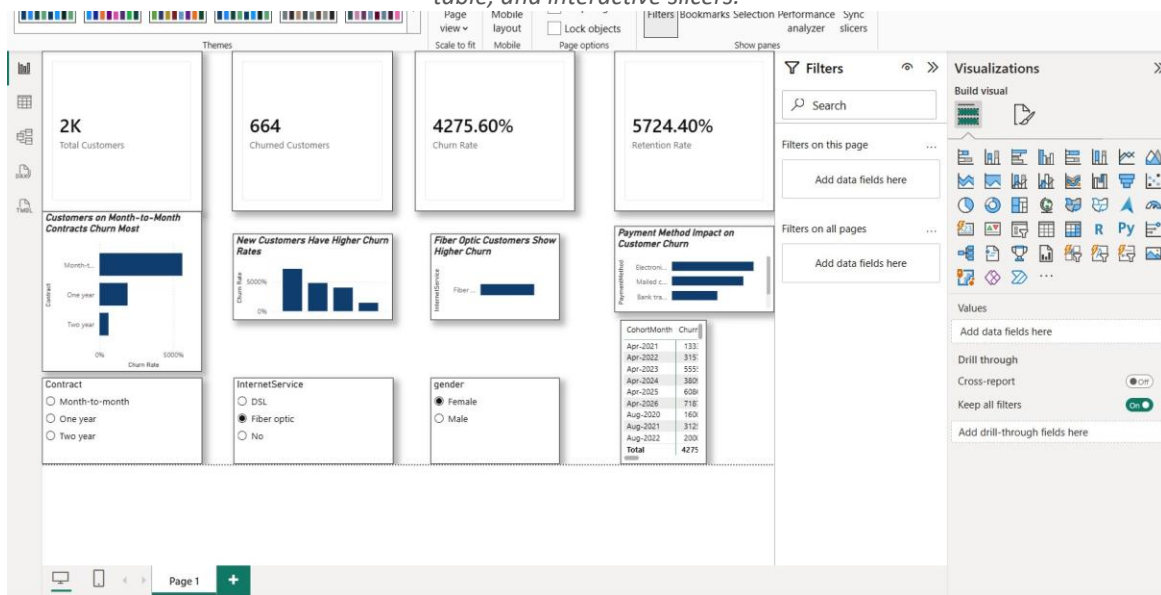


Figure 6. Dashboard filtered by Female + Fiber Optic customers, showing how slicers dynamically update all visuals simultaneously.

Churn by Contract Type

Contract type is the single strongest predictor of churn in this dataset. The pattern is stark and consistent:

Contract Type	Churn Rate	Risk Level
Month-to-month	~42%	HIGH
One year	~11%	MEDIUM
Two year	~3%	LOW

Month-to-month customers churn at roughly 14x the rate of two-year contract customers. This is the clearest single lever the business can pull: every month-to-month customer successfully converted to a one- or two-year contract is a customer whose churn probability drops dramatically. The visual shows this as three horizontal bars of vastly different lengths, with month-to-month being far longer than one year, which is far longer than two year.

Churn by Tenure Group

The tenure group analysis reveals when customers are most vulnerable to leaving. New customers in their first 12 months show the highest churn rate, which drops significantly as customers reach their second year and beyond. This early-exit pattern is a hallmark of poor onboarding or unmet expectations during the initial service experience.

Tenure Group	Churn Rate	Interpretation
0–12 Months	Highest	Critical onboarding window
13–24 Months	Moderate	Settling period
25–48 Months	Lower	Engaged customers
49+ Months	Lowest	Loyal long-term base

The business needs to focus retention energy on the first 12 months. A customer who survives past their first year is significantly more likely to stay long-term. Every resource invested in onboarding, proactive support, and early engagement in month 1–12 has outsized returns compared to retention efforts aimed at longer-tenured customers.

Churn by Internet Service

Internet service type shows a clear two-tier split in churn behavior. Fiber optic customers churn at a substantially higher rate than DSL customers, despite (or possibly because of) fiber optic being the premium, higher-cost option. This is a warning sign worth investigating: customers paying more for the premium service are less satisfied or less retained than those on the basic tier. Possible causes include higher price sensitivity, greater competition in the fiber market, or a gap between fiber optic marketing

promises and actual service experience. Customers with no internet service show the lowest churn of all three groups.

Churn by Payment Method

Payment method reveals a behavioral pattern strongly associated with churn. Electronic check users churn at the highest rate of any payment method, followed by mailed check, while customers on automatic payment methods (bank transfer or credit card automatic) churn at significantly lower rates. The automatic payment finding is intuitive: customers who set up automatic billing have made an active commitment to staying, and the friction of cancellation is higher. Electronic check users, by contrast, are making a conscious payment each billing cycle — every payment is a decision point where they could choose to stop.

Payment Method	Churn Risk	Action
Electronic check	Highest	Priority for auto-pay migration
Mailed check	High	Offer auto-pay incentive
Bank transfer (auto)	Low	Maintain & replicate
Credit card (auto)	Lowest	Maintain & replicate

Cohort Analysis

The CohortMonth matrix groups customers by their simulated signup month and shows how churn accumulates across cohorts. Earlier cohorts (further back in time) naturally show higher total churn counts simply because those customers have had more time to churn. The key insight from cohort analysis is identifying whether newer cohorts are churning at faster or slower rates than older ones — a rising rate across recent cohorts would signal a worsening retention problem, while a stable or falling rate suggests retention efforts are working. The dashboard slicer functionality allows filtering the cohort table by contract type, internet service, and gender to drill into specific segment behavior.

Key Insights & Recommendations

1. Prioritize converting month-to-month customers to annual contracts. This is the single highest-impact action available. Offer a meaningful incentive — a first-month discount, a bundled service upgrade, or a loyalty reward — specifically targeted at month-to-month customers who have been active for 3–6 months. Even converting 20% of month-to-month customers to one-year contracts would reduce overall churn rate substantially.
2. Build a dedicated 90-day onboarding program for new customers. Since churn is highest in the first 12 months, proactive outreach (welcome calls, setup support, usage tips, check-in emails at 30/60/90 days) specifically targeted at new customers can dramatically reduce early exits. Consider assigning a dedicated support contact for the first 3 months.

3. Investigate and fix the fiber optic churn problem. Premium customers should be the most retained, not the most likely to leave. Conduct a survey or exit interview analysis specifically for churned fiber optic customers to identify whether the issue is price, service quality, competition, or expectation mismatch — then fix it before acquiring more fiber optic customers.
4. Run an automatic payment migration campaign. Target all electronic check and mailed check customers with an incentive to switch to bank transfer or credit card automatic payment. A one-time bill credit of \$5–10 for switching is likely to pay for itself many times over through improved retention. Automatic payment customers churn at much lower rates, making this a high-return, low-cost intervention.
5. Create a high-risk customer early-warning system. The highest-risk customer profile in this dataset is: month-to-month contract + fiber optic internet + electronic check payment + tenure under 12 months. Flag every customer matching this profile in the CRM and assign proactive retention outreach within their first 60 days.
6. Use cohort tracking as an ongoing KPI. Set up a monthly cohort churn report in Power BI so the team can see whether each new cohort of customers is churning faster or slower than prior cohorts. This turns churn from a lagging indicator into an early-warning system, allowing retention campaigns to be deployed before churn fully materializes.

Tools & Deliverables

- Analysis tool: Microsoft Power BI Desktop — Power Query, DAX measures, interactive KPI dashboard with slicers.
- Dataset: Telco Customer Churn Dataset, sourced from Kaggle (7,043 customers, 21 fields).
- Deliverable 1: this report (insights and recommendations).
- Deliverable 2: the Power BI workbook (Customer_Churn_Analysis.pbix) containing the cleaned data model, all DAX measures, and the interactive dashboard.
- Deliverable 3: a GitHub repository documenting the project with README, dashboard screenshots, and a link to the .pbix file or a screen recording walkthrough.